



DEPARTMENT OF CSE-AIML

Weekly Monitoring Report
Academic Year 2026-27

Project Monitoring Report: Week 1

Date: 09/10/2026

Project Details

Project Title:

Machine Learning-Based In-Ovo Chick Embryo Sex Prediction Using Egg Morphology

Group Members:

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Faculty Guide: Ms. Hafsa Tasneem

Weekly Progress Update

1. Tasks Completed:

- Developed the smartphone-based egg imaging setup with controlled lighting.
- Implemented egg image processing and morphological feature extraction.
- Extracted length, width, area, eccentricity and extent.
- Tested KNN, Boosted Trees, Cubic SVM and Wide Neural Network models.
- Wide Neural Network achieved a maximum accuracy of **88.9%** and mean accuracy of **81.5%**.

2. Challenges Faced:

- Limited dataset of **121 viable embryos**.
- Variation in model performance across trials.
- Need for consistent egg positioning and image conditions.

3. Planned Tasks for Next Week:

- Further validate the models and results.
- Improve image processing and prediction accuracy.
- Complete project documentation and analysis.

4. Faculty Guide's Feedback & Suggestions:

- Improve model reliability through further testing.
- Clearly document the dataset, methodology and results.
- Focus on reducing classification errors.

Remarks:

Initial results demonstrate the feasibility of using egg morphology and machine learning for non-invasive embryo sex prediction. Further validation is required.

Student Representative Signature: _____

Faculty Guide Signature: _____

HOD/Coordinator Signature: _____

Instructions:

- This report must be submitted weekly to the assigned faculty guide.
- Ensure timely updates and address any feedback given by the guide.
- Continuous assessment will be based on these reports.

Coordinator

HOD